

Homework 8: Diagonalization

M. Usman Rashid

BCS A, E, F, G

Problem. The Fibonacci sequence is the sequence of positive integers defined recursively via

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}, \quad \forall, \quad n \geq 1.$$

The first few terms of this sequence are 0,1,1,2,3,5,8,13,21,34, Find an explicit formula for F_n that does not depend on the previous terms in the sequence.

Remark. The eigenvalues for this problem are known to be as “*Golden Ratio*”.

Reference Link. <https://mathworld.wolfram.com/GoldenRatio.html>

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Good Luck